The Egg Drop Problem: 1-3 vs 2-2 Approach

A Mathematical Analysis

Abstract

We analyze two distinct strategies for the classic Egg Drop Problem with 4 eggs and N floors: the "1-3 approach" (the standard optimal recursive strategy) and the "2-2 approach" (a structural grouping strategy). We will derive the asymptotic number of drops required for each strategy as $N \to \infty$ and compare their leading constants to determine which approach is superior.

1 Introduction and Preliminaries

In the classic Egg Drop Problem, we are given e eggs and a building with N floors. We wish to find the lowest floor from which an egg will break when dropped, using the minimum possible number of drops in the worst-case scenario.

Let $D_e(N)$ be the minimum number of drops required. It is a well-known result that the maximum number of floors N we can test with e eggs and D drops is given by the sum of binomial coefficients:

$$N(D, e) = \sum_{k=1}^{e} \binom{D}{k} \tag{1}$$

For a fixed number of eggs e and large D, the dominating term of this sum is $\binom{D}{e} \approx \frac{D^e}{e!}$. Setting $\frac{D^e}{e!} \approx N$, we find the asymptotic behavior of the optimal number of drops:

$$D_e(N) \approx (e!\, N)^{1/e} \tag{2}$$

For $e = 2$, $D_2(N) \approx (2N)^{1/2} = \sqrt{2N}$.
For $e = 3$, $D_3(N) \approx (6N)^{1/3}$.

2 The 1-3 Approach

The "1-3 approach" refers to the standard optimal strategy for 4 eggs. We drop the first egg at carefully chosen intervals. If it breaks, we are left with 3 eggs to search the intermediate floors, effectively reducing the problem to the 3-egg case.

Using the general asymptotic formula for $e = 4$, the maximum number of floors that can be covered with D drops is:

$$N \approx \frac{D^4}{4!} = \frac{D^4}{24} \tag{3}$$

Solving for D, we find the number of drops required for the 1-3 approach:

$$D_{1-3}(N) \approx (24N)^{1/4} = 24^{1/4} N^{1/4} \tag{4}$$

The constant factor here is $24^{1/4} \approx 2.213$.

3 The 2-2 Approach

In the "2-2 approach", we structure our search differently. We partition the N floors into blocks of size F, creating N/F "superfloors". We allocate our 4 eggs into two pairs:

1. **Phase 1:** We use the first 2 eggs to determine which superfloor contains the critical floor. This is equivalent to solving the 2-egg problem for N/F items.
2. **Phase 2:** Once the critical superfloor is identified (and our first 2 eggs are potentially broken), we use the remaining 2 eggs to search the F individual floors within that superfloor.

Using the asymptotic formula for 2 eggs, Phase 1 requires approximately $\sqrt{2(N/F)}$ drops, and Phase 2 requires approximately $\sqrt{2F}$ drops. The total number of drops is:

$$D_{2-2}(F) \approx \sqrt{\frac{2N}{F}} + \sqrt{2F} = \sqrt{2N}F^{-1/2} + \sqrt{2}F^{1/2} \tag{5}$$

To optimize this strategy, we must choose F to minimize $D_{2-2}(F)$. We take the derivative with respect to F and set it to zero:

$$\frac{d}{dF}\left(\sqrt{2N}F^{-1/2} + \sqrt{2}F^{1/2}\right) = -\frac{1}{2}\sqrt{2N}F^{-3/2} + \frac{1}{2}\sqrt{2}F^{-1/2} = 0 \tag{6}$$

Multiplying through by $2F^{3/2}$, we obtain:

$$-\sqrt{2N} + \sqrt{2}F = 0 \implies F = \sqrt{N} \tag{7}$$

Substituting $F = \sqrt{N}$ back into our total drops equation yields the optimized drops for the 2-2 approach:

$$D_{2-2}(N) \approx \sqrt{\frac{2N}{\sqrt{N}}} + \sqrt{2\sqrt{N}} = \sqrt{2}N^{1/4} + \sqrt{2}N^{1/4} = 2\sqrt{2}N^{1/4} = \sqrt{8}N^{1/4} \tag{8}$$

The constant factor here is $\sqrt{8} = 64^{1/4} \approx 2.828$.

4 Conclusion

Both strategies require a number of drops on the order of $N^{1/4}$. However, the leading constants differ significantly:

- **1-3 Approach:** Drops $\approx 24^{1/4}N^{1/4} \approx 2.213N^{1/4}$
- **2-2 Approach:** Drops $\approx \sqrt{8}N^{1/4} \approx 2.828N^{1/4}$

Final Verdict

The **1-3 approach** is mathematically superior to the 2-2 approach. By allowing the first egg to establish an optimal variable partition rather than forcing a strict grid of superfloors, the 1-3 strategy takes full advantage of the generalized $\binom{D}{4}$ combinatorial bound, resulting in a roughly 22% reduction in the number of required drops for large N.